On Upper-Confidence Bound Policies for Switching Bandit Problems

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Abstract. Many problems, such as cognitive radio, parameter control of a scanning tunnelling microscope or internet advertisement, can be modelled as non-stationary bandit problems where the distributions of rewards changes abruptly at unknown time instants. In this paper, we analyze two algorithms designed for solving this issue: *discounted UCB* (D-UCB) and *sliding-window UCB* (SW-UCB). We establish an upper-bound for the expected regret by upper-bounding the expectation of the number of times suboptimal arms are played. The proof relies on an interesting Hoeffding type inequality for self normalized deviations with a random number of summands. We establish a lower-bound for the regret in presence of abrupt changes in the arms reward distributions. We show that the discounted UCB and the sliding-window UCB both match the lower-bound up to a logarithmic factor. Numerical simulations show that D-UCB and SW-UCB perform significantly better than existing soft-max methods like EXP3.S.

1 Introduction

Multi-armed bandit (MAB) problems, modelling allocation issues under uncertainty, are fundamental to stochastic decision theory. The archetypal MAB problem may be stated as follows: there is a bandit with K independent arms. At each time step, the agent chooses one arm and receives a reward accordingly. In the stationary case, the distribution of the rewards are initially unknown, but are assumed to remain constant during all games. The agent aims at minimizing the expected *regret* over T rounds, which is defined as the expectation of the difference between the total reward obtained by playing the best arm and the total reward obtained by using the algorithm. For several algorithms in the literature (e.g. [20, 1]), as the number of plays T tends to infinity, the expected total reward asymptotically approaches that of playing a policy with the highest expected reward, and the regret grows as the logarithm of T. More recently, finite-time bounds for the regret and improvements have been derived (see [5, 2, 16]), but those improvements do not address the issue of non-stationarity.

Though the stationary formulation of the MAB allows to address exploration versus exploitation challenges in a intuitive and elegant way, it may fail to be adequate to model an evolving environment where the reward distributions undergo changes in time. As an example, in the cognitive medium radio access

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problem [19], a user wishes to opportunistically exploit the availability of an empty channel in a multiple channel system; the reward is the availability of the channel, whose distribution is unknown to the user. Another application is real-time optimization of websites by targetting relevant content at individuals, and maximizing the general interest by learning and serving the most popular content (such situations have been considered in the recent Exploration versus Exploitation (EvE) PASCAL challenge by [14], see also [18] and the references therein). These examples illustrate the limitations of the stationary MAB models. The probability that a given channel is available is likely to change in time. The news stories a visitor of a website is most likely to be interested in vary in time.

To model such situations, non-stationary MAB problems have been considered (see [17, 14, 22, 24]), where distributions of rewards may change in time. Motivated by the problems cited above, and following a paradigm widely used in the change-point detection literature (see [12, 21] and references therein), we focus on non-stationary environments where the distributions of the rewards undergo abrupt changes. We show in the following that, as expected, policies tailored for the stationary case fail to track changes of the best arm.

Section 2 contains the formal presentation of the non-stationary setting we consider, together with two algorithms adressing this exploration/exploitation dilemma : D-UCB and SW-UCB. D-UCB had been proposed in [17] with empirical evidence of efficiency, but no theoretical analysis. SW-UCB is a new UCB-like algorithm that appears to perform slightly better in switching environments. In Section 3, we provide upper-bounds on the performance of D-UCB and SW-UCB; moreover, we provide a lower-bound on the performance of any algorithm in abruptly changing environments, that almost matches the upper-bounds. As a by-product, we show that any policy (like UCB-1) that achieves a logarithmic regret in the stationary case cannot reach a regret of order smaller than $T/\log(T)$ in the presence of switches. D-UCB is analyzed in Section 4; it relies on a novel deviation inequality for self-normalized averages with random number of summands which is stated in Section 7 together with some technical results. A lower bound on the regret of any algorithm in an abruptly changing environment is given in Section 5. In Section 6, two simple Monte-Carlo experiments are presented to support our findings.

2 Algorithms

In the sequel, we assume that the set of arms is $\{1,\dots,K\}$, and that the rewards $\{X_t(i)\}_{t\geq 1}$ for arm $i \in \{1,\dots,K\}$ are modeled by a sequence of independent random variables from potentially different distributions (unknown to the user) which may vary across time but remain bounded by $B > 0$. For each $t > 0$, we denote by $\mu_t(i)$ the expectation of the reward $X_t(i)$ for arm i. Let i_t^* be the arm with highest expected reward at time t (in case of ties, let i_t^* be one of the arms with highest expected rewards). The regret of a policy π is defined as the expected difference between the total rewards collected by the optimal policy